TOPOLOGICAL FINITE TYPENESS OVER AN ADMISSIBLE RING IS TESTED BY A SINGLE IDEAL OF DEFINITION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a commutative diagram with exact rows; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The preceding equivalence of four conditions for a map of complete weakly admissible rings. Same chapter, Lemma lemma-quotient-restricted-power-series, referred to in the statement at lines 6660-6661. Fixes the conditions that the new one is compared with.
- Admissible and adic rings, ideals of definition and weak ideals of definition. Same chapter, the definitions of these notions, used at proof lines 6673-6677. Supply the containment of a power of the chosen ideal in an arbitrary weak ideal of definition.
- Finiteness of type descends through a nilpotent ideal. Algebra, Lemma algebra-lemma-finite-type-mod-nilpotent, invoked at proof lines 6689-6692. Gives that the map of quotients is of finite type once the relevant kernel is nilpotent.

Source reference. Original lemma 0G8U: sources/stacks-chow/tags.txt, source lines 20279–20279. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ANU: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 6560–6560. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *In Lemma 0ANU if B is admissible (for example adic), then the equivalent conditions (a) – (d) are also equivalent to*

(e) *$B \to A$ is taut and $B/J \to A/I$ is of finite type for some ideal of definition $J \subset B$ where $I \subset A$ is the closure of JA.*

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Proof. It is enough to show that (e) implies (a). Let $J' \subset B$ be a weak ideal of definition and let $I' \subset A$ be the closure of $J'A$. We have to show that $B/J' \to A/I'$ is of finite type. If the corresponding statement holds for the smaller weak ideal of definition $J'' = J' \cap J$, then it holds for J'. Thus we may assume $J' \subset J$. As J is an ideal of definition (and not just a weak ideal of definition), we get $J^n \subset J'$ for some $n \geq 1$. Thus we can consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I/I' & \longrightarrow & A/I' & \longrightarrow & A/I & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & J/J' & \longrightarrow & B/J' & \longrightarrow & B/J & \longrightarrow & 0 \end{array}$$

with exact rows. Since $I' \subset A$ is open and since I is the closure of JA we see that $I/I' = (J/J') \cdot A/I'$. Because J/J' is a nilpotent ideal and as $B/J \to A/I$ is of finite type, we conclude from Algebra, Lemma 0G8U that A/I' is of finite type over B/J' as desired. $\square$